

NIFTY 100

Data Foundation Analyst's Guide

Clustering • Portfolio Analytics • REST API • Dashboard
Sprint 6 deliverable — version 0.6.0

Prepared for internal equity-research workflows over the Nifty 100 universe of 92 companies with complete financial datasets.

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1. Introduction and scope

This guide documents the analytics and API layer delivered in Sprint 6. It describes how the 92-company Nifty 100 universe is clustered into five investment archetypes, how portfolio and outlier statistics are produced, and how to consume the same data through a versioned REST API, a Streamlit dashboard and PDF reports.

The full stack runs on a single SQLite database with 16 live endpoints under /api/v1. Every number quoted in this guide is computed directly from that database and can be reproduced by the scripts referenced in each section.

Audience: equity analysts, quant researchers and data engineers who need a repeatable, documented pipeline from raw annual reports to decision-ready screens.

2. Data foundation

The database (db/nifty100.db) holds ten core tables. Row counts are reported by the health endpoint and are pinned by the acceptance gates:

Table	Rows	Granularity
companies	92	one per Nifty 100 member with profile metadata
profitandloss	1,057	company x fiscal year
balancesheet	1,140	company x fiscal year
cashflow	1,056	company x fiscal year
financial_ratios	1,155	company x fiscal year KPI sheet
market_cap	552	company x calendar year (2019-2024)
sectors	92	sector mapping incl. market-cap category
peer_groups	56	peer-group membership and benchmarks
peer_percentiles	—	per-metric percentile ranks in groups
documents	1,456	annual-report registry

Fourteen data-quality rules (DQ-01 to DQ-14) are enforced by the ETL validator: primary-key uniqueness, company+year uniqueness, foreign-key integrity, the balance-sheet equation, OPM cross-checks, positive sales, required columns, null pk/year checks, numeric typing, duplicate rows, year range, URL format and negative-value checks. Every rule has a dedicated unit test.

3. Clustering methodology

Five features capture profitability, leverage and growth:

- return_on_equity_pct
- debt_to_equity
- revenue_cagr_5yr
- fcf_cagr_5yr
- operating_profit_margin_pct

Missing values are imputed with the sector median of the feature, then the feature matrix is standardised with StandardScaler before KMeans (k=5, random_state=42, k-means++ initialisation). The elbow plot (reports/elbow_plot.png) supports a five-cluster solution. The pipeline is implemented in src/analytics/clustering.py.

4. Cluster profiles

Five archetypes emerge across the universe. Mean and median values per feature:

Cluster	Count	ROE med	D/E med	Rev CAGR med	FCF CAGR med	OPM med
0 (High-Quality Compounders)	60	17.0	0.24	10.6	12.1	19.1
1 (High-Margin Franchises)	14	14.7	0.07	13.3	11.8	85.1
2 (Defense High-ROE Leaders)	2	4280.3	0.56	9.7	32.4	28.5
3 (Leveraged Financials)	15	15.8	6.81	18.2	-6.3	38.6
4 (Cash-Flow Outliers)	1	15.6	0.02	9.5	228.8	75.6

Reading the archetypes: cluster 0 is the broad quality compounder core; cluster 1 contains insurers and franchises with very high operating margins; cluster 2 is a two-member defence pairing (BEL, HAL) with extreme ROE; cluster 3 is the leveraged financials cohort (banks, NBFCs, ADANIGREEN); cluster 4 is a single cash-flow outlier (CIPLA).

4.1 Cluster membership

Cluster 0 — High-Quality Compounders (60 companies):

RELIANCE, SHREECEM, MARUTI, TCS, GAIL, ASIANPAINT, INFY, PIDILITIND, HCLTECH, NTPC, EICHERMOT, BPCL, APOLLOHOSP, NESTLEIND, IOC, TATAMOTORS, ONGC, TATASTEEL, TECHM, JSWSTEEL, MOTHERSON, ABB, BAJAJ-AUTO, GODREJCP, TORNTPHARM, DIVISLAB, LT, GRASIM, M&M, TATAPOWER, NAUKRI, DABUR, ITC, HAVELLS, BOSCHLTD, AMBUJACEM, ADANIPOWER, SUNPHARMA, TVSMOTOR, DRREDDY, ADANIENSOL, DMART, IRCTC, ADANIENT, ATGL, JINDALSTEL, JSWENERGY, SIEMENS, TITAN, BRITANNIA, TATACONSUM, SBIN, SBILIFE, NHPC, LICI, LODHA, BHEL, DLF, LTIM, TRENT

Cluster 1 — High-Margin Franchises (14 companies):

HINDALCO, HINDUNILVR, ICICIGI, JIOFIN, COALINDIA, BHARTIARTL, HDFCLIFE, KOTAKBANK, ADANIPORTS, POWERGRID, HEROMOTOCO, ICICIPRULI, INDIGO, BAJAJHLDNG

Cluster 2 — Defense High-ROE Leaders (2 companies):

BEL, HAL

Cluster 3 — Leveraged Financials (15 companies):

HDFCBANK, ICICIBANK, INDUSINDBK, AXISBANK, CHOLAFIN, BAJAJFINSV, PFC, PNB, SHRIRAMFIN, BAJFINANCE, BANKBARODA, IRFC, RECLTD, ADANIGREEN, CANBK

Cluster 4 — Cash-Flow Outliers (1 companies):

CIPLA

5. Cross-sectional findings

The correlation heatmap (reports/correlation_heatmap.png) shows pairwise Spearman correlations between ten headline KPIs. Per-sector Z-score analysis flags companies that sit more than three standard deviations from their sector peers:

Company	Sector	Metric	Value	Z-score
INDIGO	Consumer Discretionary	return_on_equity_pct	892.57	3.60
INDIGO	Consumer Discretionary	return_on_capital_employed_pct	5,643.45	3.60
INDIGO	Consumer Discretionary	return_on_assets_pct	668.33	3.60
TATAMOTORS	Consumer Discretionary	free_cash_flow_cr	45,133.00	3.35
ADANIGREEN	Energy	debt_to_equity	6.60	3.26
LICI	Financials	return_on_equity_pct	49.45	3.62
ICICIPRULI	Financials	return_on_capital_employed_pct	754.89	3.30
BAJAJHLDNG	Financials	net_profit_margin_pct	432.73	4.49
BAJAJHLDNG	Financials	return_on_assets_pct	11.30	4.06
BAJAJHLDNG	Financials	interest_coverage	3,764.50	3.67

Flags are concentrated in Consumer Discretionary (INDIGO, TATAMOTORS) and Financials (LICI, ICICIPRULI, BAJAJHLDNG) where the median profile is distorted by structurally different businesses; ADANIGREEN is flagged on leverage (D/E 6.6x). These points should be reviewed before using the median as a sector anchor.

6. Portfolio statistics

Population percentiles (P10-P90), mean and standard deviation for the ten headline KPIs across the universe:

KPI	n	P10	P25	P50	P75	P90	Mean	Std
ROE %	90	8.4	11.6	16.3	25.4	45.5	124.6	640.7
ROCE %	90	4.3	11.3	20.6	36.2	121.0	197.5	840.6
NPM %	91	4.6	7.2	14.3	20.4	29.3	20.1	45.4
OPM %	91	9.8	16.0	25.8	40.2	84.0	34.7	27.5
ROA %	90	1.1	2.4	6.2	14.0	22.6	62.3	319.5
D/E	91	0.0	0.1	0.4	1.6	6.4	1.7	2.9
Interest coverage	91	1.6	3.2	11.4	58.1	192.3	126.1	485.9
Revenue CAGR %	90	5.8	8.4	11.8	17.2	21.1	12.9	7.4
PAT CAGR %	84	4.4	9.9	16.3	23.8	39.3	21.1	21.5
FCF (cr)	90	-14,847.2	-1,340.0	1,068.5	8,164.0	20,818.1	706.2	23,312.3

7. REST API reference

FastAPI serves 16 endpoints under `/api/v1` with CORS open to all origins, request logging and a health check that reports row counts:

Method	Path	Purpose
GET	<code>/health</code>	status, version, uptime, DB row counts
GET	<code>/companies</code>	list with sector/market-cap/search filters
GET	<code>/companies/{ticker}</code>	full profile + latest KPIs (404 if unknown)
GET	<code>/companies/{ticker}/pl</code>	P&L history, optional from/to year
GET	<code>/companies/{ticker}/bs</code>	balance-sheet history
GET	<code>/companies/{ticker}/cashflow</code>	cash-flow history
GET	<code>/companies/{ticker}/ratios</code>	KPI ratios, optional year
GET	<code>/companies/{ticker}/tearsheet</code>	pre-generated PDF
GET	<code>/screener</code>	threshold screen (cursors 400 on bad values)
GET	<code>/sectors</code>	sector aggregates and market-cap splits
GET	<code>/sectors/{sector}/companies</code>	companies in a sector
GET	<code>/peers/{group_name}</code>	peer membership + percentile ranks
GET	<code>/companies/{ticker}/peers/compare</code>	8-axis radar data
GET	<code>/market-cap/{ticker}</code>	2019-2024 valuation multiples
GET	<code>/portfolio/stats</code>	P10-P90 portfolio statistics
GET	<code>/companies/{ticker}/documents</code>	annual-report registry

Machine-readable contracts are exported to `docs/openapi.json` and a Postman collection at `docs/nifty100.postman_collection.json`. Run the API with: `uvicorn src.api.main:app --port 8000`

8. Screener and presets

GET /api/v1/screener applies threshold filters on one frame per company at its latest screening year and sorts by the composite quality score. Supported filters:

- min_roe
- max_de
- min_fcf
- sector
- min_rev_cagr_5yr
- min_pat_cagr_5yr
- max_pe

Example: curl "/api/v1/screener?min_roe=15" returns 53 companies with ROE $\geq$ 15%. Non-numeric parameters return HTTP 400. The Streamlit dashboard exposes the same screen with six one-click presets (Quality, Value, Growth, Dividend, Debt-Free, Turnaround); the API returns the same company set for equal thresholds (verified by an integration test).

9. Dashboard guide

Run: `streamlit run src/dashboard/app.py` (port 8501 by default). The dashboard has eight pages: Home, Company Profile, Screener, Peers, Trends, Sectors, Capital Allocation and Reports. Company Profile resolves any ticker to a profile card, six KPI tiles, revenue/profit bars, ROE/ROCE trends and pros/cons. Data is cached for 10 minutes; per-ticker company-profile loads complete in under 3 seconds (measured, `output/perf_notes.md`).

10. Known limitations and deviations

- The live screener considers latest-year values; companies with negative or zero revenue in the base year have undefined CAGR flags and are excluded from that feature.
- The database carries 10 distinct broad sectors, not the 11 the planning documents assumed; API and tests follow the data.
- Four tearsheets (ATGL, JIOFIN, PNB, SBIN) could not be generated in Sprint 5 (missing 10-year series) and are documented in `output/skipped_tearsheets.csv`; the tearsheet endpoint 404s for them.
- Z-score outliers are data-driven and flagged, not corrected; sector medians still include them unless the analyst filters.

Every deviation is mirrored in `docs/sprint6_review.md` and in the acceptance checklist.

11. Quality assurance and sign-off

The Sprint 6 verification harness (`scripts/verify_sprint6.py`) runs 20 acceptance gates AC-01 to AC-20, including: 92 companies clustered to 5 labelled archetypes; 16 live endpoints with correct status codes; the screener matching the dashboard; 10 concurrent API calls finishing inside 10 seconds; dashboard profile loads under 3 seconds; 381 pytest tests with zero failures (`reports/pytest_report.html`); 23 deliverables archived in `output/final_deliverables/`. The acceptance checklist (`docs/acceptance_checklist.pdf`) records a PASS/FAIL per gate and the day-45 sign-off. This guide, the API and the clustering pipeline together close Epics 10, 11 and 12.